

Memory-Pruning Algorithm for Gaussian Processes Regression

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Outline

- 1 Scientific Context (Neurostimulation, data collection, objectives...)
- 2 Method Description (Gaussian Process Regression, Memory Pruning...)
- 3 Key Results (Benchmarks, neurostimulation...)
- 4 Perspectives

Neurostimulation

Data Collection

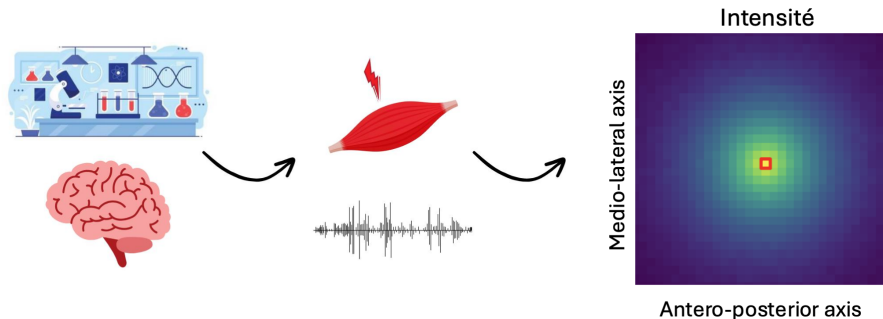


Figure: Data collection process

Feature Name	Description	Type
x	position in matrix	Integer
y	position in matrix	Integer
τ	stimulation duration	Numerical
ω	frequency	Numerical
I	intensity	Numerical

...

Summary of the main features in the dataset.

Gaussian Processes Regression to find the best parameters set.

Advantages:

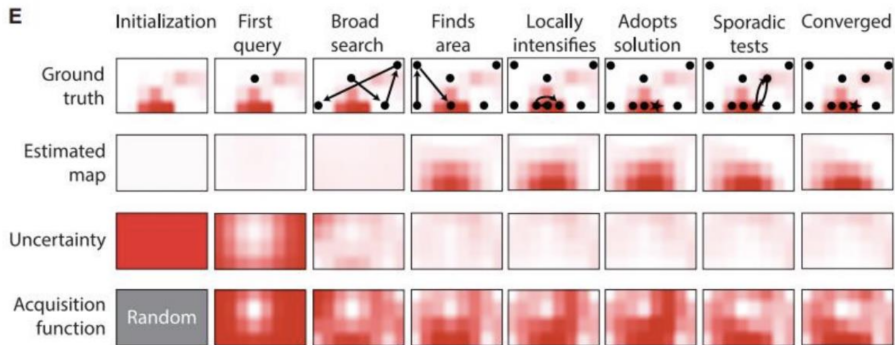
- Requires little data
- Can be done sequentially

Drawbacks:

- Poor performance in high-dimensional settings
- Very demanding in time and memory

⇒ **Obstacle to miniaturisation**

Bayesian Optimisation (BO) with Gaussian Processes (GP)



From Bonizzato et al., 2023, Cell Reports Medicine 4, 101008

Figure: Decomposition of the search in BO.

GP-BO: Memory and Computation Time

Update: Need to store and invert the kernel matrix of all collected data points...

- $N \times N$ matrix: N^2 for storage
- $\mathcal{O}(N^3)$ operations for inversion

$\implies$ Obvious limitation if search space grows

Idea: Only keep m memory points at update step

Online Paging Problem:

- A small-sized fast memory
- An infinite slow memory

→ Randomly removing elements from memory works well on average

Algorithm 2 Bayesian Optimization with Memory Pruning (MP-BO)

```
1: Init: Randomly sample a point  $\mathbf{x}_1$  and its response  $y_1$ .  $\mathcal{D}_{1:1} := \{(\mathbf{x}_1, y_1)\}$ 
2: for  $n = 1, 2, \dots$  do
3:    $\mathbf{x}_{n+1} = \arg \max_{\mathbf{x}} AF(\mathbf{x} | \mathcal{D}_{1:n})$  ▷ Find new  $\mathbf{x}_{n+1}$  to sample
4:    $y_{n+1} = f(\mathbf{x}_{n+1}) + \varepsilon_{n+1}$  ▷ Sample the objective function
5:   if  $n \geq q^*$  then
6:      $(\tilde{\mathbf{x}}, \tilde{y}) = u(\mathcal{D}_{1:n})$  ▷ Find a query to delete and remove it from training set
7:      $\mathcal{D}_{1:n} = \mathcal{D}_{1:n} \setminus \{(\tilde{\mathbf{x}}, \tilde{y})\}$ 
8:   end if
9:    $\mathcal{D}_{1:n+1} = \mathcal{D}_{1:n} \cup \{(\mathbf{x}_{n+1}, y_{n+1})\}$  ▷ Augment the data set and update the surrogate model
10: end for
```

Optimization Time

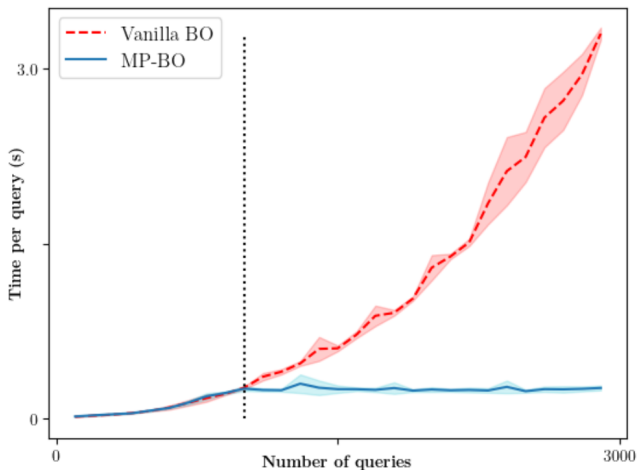


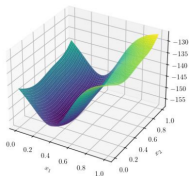
Figure: Bounded time of update with MPBO.

Benchmarks

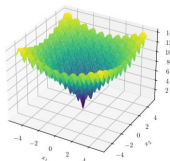
Frameworks

- Gpytorch
- BOtorch

Hartmann 6D (projection 2D)



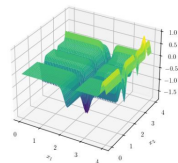
Ackley (2D)



Datasets

- Fonctions synthétiques
- Neurostimulation sur NHP

Michalewicz (2D)



Algorithmes

- Vanilla BO
- MP-BO

Evaluation

- Regret

MP-BO: Results

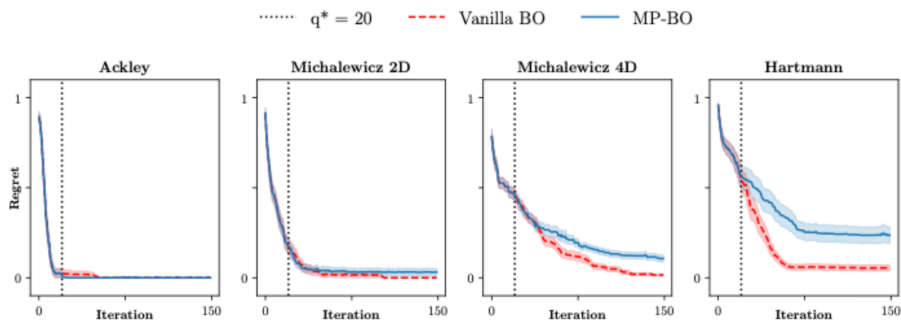


Figure: Can MP-BO achieve low regret ?

MP-BO: Results

--- Vanilla BO at q^* — MP-BO after 150 iterations

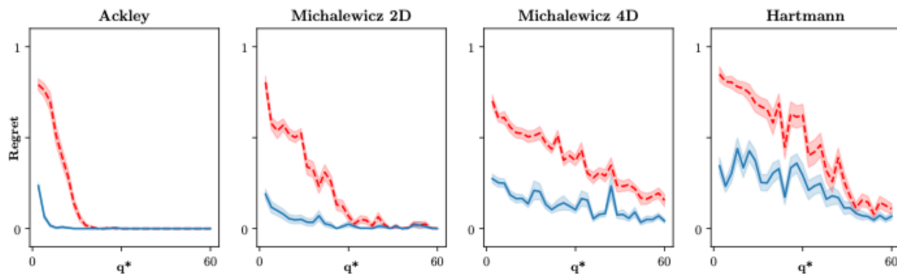


Figure: Can MP-BO learn after fixing the memory size ?

MP-BO: Results

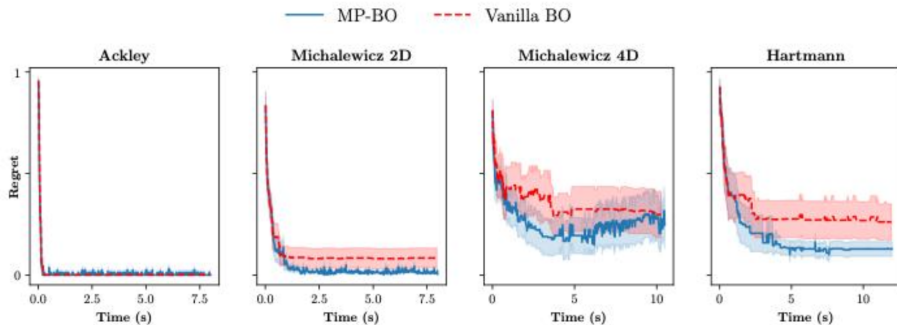


Figure: What is the benefice in time ?

MP-BO: Results

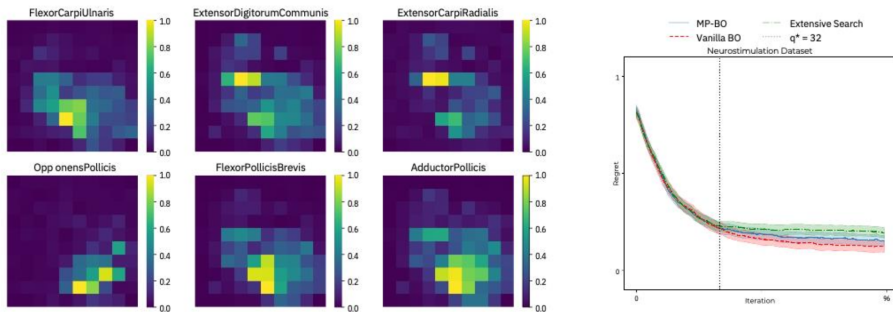


Figure: Results on NHP neurostimulation datasets.

- Compare with other existing methods (SpGP, dimensionality reduction)
- Test more complex strategies

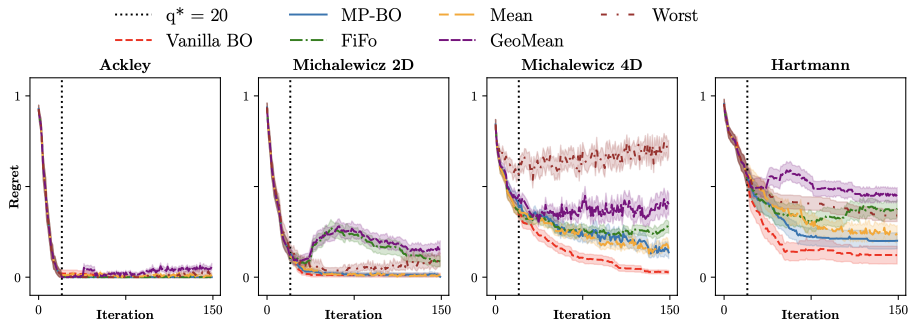


Figure: Results with other data selection.